

Practice Question Sets

Optional Computer Science — Class X — First Terminal Examination 2083

Contains 3 full practice sets, each followed by complete solutions.

Sub code '3031' • Full Marks: 50 • Time: 2 hrs (per set)

Symbol No. _____

Sub code '3031'

Practice Question — Set 1

First Terminal Examination - 2083

Subject: Optional Computer Science

Class: X Group 'A'

Full Marks: 50 Time: 2 hrs

Group 'A'

Rewrite the correct option in your answer sheet. [10 × 1 = 10]

1. The connection pattern of computers in a network is called...

- (a) Protocol (b) Topology (c) Structure (d) IRC

2. What is the primary purpose of a primary key in a database table?

- (a) To store only text data (b) To uniquely identify each record in a table (c) To combine multiple fields into one
(d) To sort the table alphabetically

3. Which function is used to take input from the user in Python?

- (a) print() (b) input() (c) output() (d) get()

4. What does a MAC address identify?

- (a) A software application (b) A network cable (c) A unique network interface of a device (d) An internet protocol

5. Which type of argument is passed to a function based on its position in the function call?

- (a) Keyword argument (b) Positional argument (c) Default argument (d) Required argument

6. When reading a CSV file using pandas read_csv(), which parameter is used to specify a custom column separator?

- (a) sep (b) index (c) header (d) columns

7. Which statement imports only a specific function 'sqrt' from the 'math' module?

- (a) import math (b) from math import sqrt (c) include math.sqrt (d) using math sqrt

8. Why is Wi-Fi considered more suitable than Optical Fiber for connecting home devices wirelessly?

- (a) It provides a wired high-speed connection (b) It allows wireless communication within a limited range (c) It is used only for broadcasting television (d) It requires physical cables

9. Which Plotly Express function is used to create a bar chart?

- (a) px.line() (b) px.bar() (c) px.pie() (d) px.scatter()

10. Which SQL command is used to remove a record from a table?

- (a) UPDATE (b) INSERT (c) DELETE (d) SELECT

Group 'B'

Write short answers to the following questions. [10 × 2 = 20]

(a) Define topology. List any two types of network topology.

(b) Write a Python turtle syntax that can draw a circle of radius 50.

(c) What is a protocol? Give one example.

(d) What is a primary key in SQL? Why is it important?

(e) Define DML with any two commands.

(f) What is transmission media? Give its two types.

(g) Differentiate between LAN and WAN with any two points.

(h) What is modulation in telecommunication?

(i) Define data analysis.

(j) What is local scope in Python? How does it differ from global scope?

Group 'C'

Write long answers to the following questions. $[4 \times 5 = 20]$

11. Answer the following questions.

a. Define database and DBMS. Also explain field and record with a suitable example. $[2+1+1]$

b. Consider the following table "Student": $[2+1+1]$

Roll_no	Name	Marks	City
1	Ram	45	Kathmandu
2	Sita	38	Janakpur
3	Hari	40	Lumbini

i. Write a SQL code to create the above table (make Roll_no the Primary key).

ii. Write a SQL code to display records of students who have scored more than 40 marks.

iii. Insert a new record (4, Gita, 30, Pokhara).

12. Answer the following questions.

a. Write a Python program to take two numbers as input and print their sum, difference, product and quotient using a function. $[1+1+2]$

b. Explain client-server architecture along with its major advantages. $[4]$

c. Define syntax error in error handling. Explain the try, except, else and finally blocks with one suitable example. $[1+3]$

— *Best of Luck* —

Solutions — Practice Set 1

Group 'A' — Answer Key

1. The connection pattern of computers in a network is called... → **(b) Topology**
2. What is the primary purpose of a primary key in a database table? → **(b) To uniquely identify each record in a table**
3. Which function is used to take input from the user in Python? → **(b) input()**
4. What does a MAC address identify? → **(c) A unique network interface of a device**
5. Which type of argument is passed to a function based on its position in the function call? → **(b) Positional argument**
6. When reading a CSV file using pandas read_csv(), which parameter is used to specify a custom column separator? → **(a) sep**
7. Which statement imports only a specific function 'sqrt' from the 'math' module? → **(b) from math import sqrt**
8. Why is Wi-Fi considered more suitable than Optical Fiber for connecting home devices wirelessly? → **(b) It allows wireless communication within a limited range**
9. Which Plotly Express function is used to create a bar chart? → **(b) px.bar()**
10. Which SQL command is used to remove a record from a table? → **(c) DELETE**

Group 'B' — Answers

(a) Topology is the physical or logical arrangement of computers and devices in a network. Two common types are Star topology and Bus topology.

(b) Write a Python turtle syntax that can draw a circle of radius 50.

```
import turtle
t = turtle.Turtle()
t.circle(50)
```

(c) A protocol is a set of rules that governs how data is formatted, transmitted, and received between devices on a network. Example: HTTP (HyperText Transfer Protocol).

(d) A primary key is a column (or set of columns) that uniquely identifies each row in a table and cannot contain NULL or duplicate values. It is important because it ensures data integrity and allows each record to be referenced uniquely, including from other tables via foreign keys.

(e) DML (Data Manipulation Language) consists of SQL commands used to manipulate data stored in tables. Two commands: INSERT and UPDATE.

(f) Transmission media is the physical or wireless path through which data travels from sender to receiver. Its two types are Guided (wired) media and Unguided (wireless) media.

(g) 1) LAN covers a small geographical area such as a single building, while WAN covers a large geographical area such as a country or continent. 2) LAN generally offers higher data transfer speed and lower cost than WAN, which needs public/leased communication links.

(h) Modulation is the process of varying a carrier signal's properties (such as amplitude, frequency, or phase) to encode digital or analog data for transmission over a communication channel.

(i) Data analysis is the process of inspecting, cleaning, transforming, and interpreting data to discover useful information, draw conclusions, and support decision-making.

(j) Local scope refers to variables defined inside a function; they can be accessed only within that function. Global scope refers to variables defined outside all functions; they can be accessed from anywhere in the program, including inside functions (unless a local variable with the same name shadows it).

Group 'C' — Answers

11. (a)

A database is an organized collection of related data stored electronically so that it can be easily accessed, managed, and updated. A DBMS (Database Management System) is software that allows users to create, store, manipulate, and retrieve data in a database (e.g., MySQL, Oracle).

A field is a single column of a table that holds one specific type of data about a record, e.g., "Name" or "Marks".

A record is a single row of a table that holds a complete set of related field values for one entity, e.g., (1, Ram, 45, Kathmandu) is one record in a student table.

11. (b) Table: Student

```
CREATE TABLE Student (  
    Roll_no INT PRIMARY KEY,  
    Name VARCHAR(30),  
    Marks INT,  
    City VARCHAR(30)  
);  
  
SELECT * FROM Student WHERE Marks > 40;  
  
INSERT INTO Student VALUES (4, 'Gita', 30, 'Pokhara');
```

12. (a)

```
def calculate(a, b):  
    print("Sum =", a + b)  
    print("Difference =", a - b)  
    print("Product =", a * b)  
    print("Quotient =", a / b)  
  
num1 = float(input("Enter first number: "))  
num2 = float(input("Enter second number: "))  
calculate(num1, num2)
```

12. (b)

Client-server architecture is a computing model in which multiple client computers request and receive services or resources from a central computer called the server, over a network. The client sends a request (e.g., a webpage or a file); the server processes the request and sends back the response.

Major advantages: 1) Centralized data management and control, making backup and security easier. 2) Resources such as files, printers, and processing power can be shared efficiently among many clients. 3) Scalability — new clients can be added without major changes to the server. 4) Easier maintenance since updates can be made centrally on the server.

12. (c)

A syntax error occurs when the code violates the grammatical rules of the programming language (e.g., a missing colon or unmatched bracket), so the program cannot even begin execution.

The try block contains code that may raise an exception. The except block catches and handles the exception if one occurs. The else block runs only if no exception occurred in the try block. The finally block always runs, whether an exception occurred or not, and is typically used for clean-up actions.

```
try:  
    num = int(input("Enter a number: "))  
    result = 10 / num
```

```
except ZeroDivisionError:
    print("Cannot divide by zero!")
except ValueError:
    print("Invalid input!")
else:
    print("Result is", result)
finally:
    print("Execution finished.")
```

Symbol No. _____

Sub code '3031'

Practice Question — Set 2

First Terminal Examination - 2083

Subject: Optional Computer Science

Class: X Group 'A'

Full Marks: 50 Time: 2 hrs

Group 'A'

Rewrite the correct option in your answer sheet. [10 × 1 = 10]

1. What do you call the physical or logical arrangement of computers in a network?

- (a) Protocol (b) Topology (c) Bandwidth (d) Domain

2. What is a foreign key used for in a database?

- (a) To uniquely identify records in the same table (b) To establish and enforce a link between two tables (c) To store multimedia data (d) To speed up printing

3. Which Python function converts a string value into an integer?

- (a) int() (b) str() (c) float() (d) input()

4. What does an IP address represent?

- (a) A hardware manufacturer (b) A device's unique network identifier (c) A software license (d) A file type

5. What type of argument has a pre-defined value that is used when no value is provided during a function call?

- (a) Positional argument (b) Keyword argument (c) Default argument (d) Required argument

6. Which pandas function is used to read data from an Excel file?

- (a) read_csv() (b) read_excel() (c) read_json() (d) read_table()

7. Which keyword is used to import an entire module in Python?

- (a) from (b) include (c) import (d) using

8. Why is optical fiber preferred for long-distance, high-speed data transmission?

- (a) It is the cheapest media available (b) It uses radio waves (c) It offers high bandwidth and low signal loss (d) It only works wirelessly

9. Which Plotly Express function is used to create a pie chart?

- (a) px.bar() (b) px.pie() (c) px.line() (d) px.scatter()

10. Which SQL command is used to retrieve records from a table?

- (a) UPDATE (b) SELECT (c) INSERT (d) DELETE

Group 'B'

Write short answers to the following questions. [10 × 2 = 20]

(a) Define network topology. Name any two types.

(b) Write a Python turtle syntax that can draw a rectangle.

(c) What is meant by data transmission speed? How is it measured?

(d) What is a foreign key? Why is it used?

(e) Define DDL with any two commands.

(f) What is guided media? Give two examples.

(g) Differentiate between wired and wireless media with two points.

(h) What is meant by a signal in telecommunication?

(i) Define data cleaning.

(j) What is the use of the global keyword in Python?

Group 'C'

Write long answers to the following questions. $[4 \times 5 = 20]$

11. Answer the following questions.

a. Define database and DBMS. Also explain primary key with a suitable example. $[2+1+1]$

b. Consider the following table “Employee”: $[2+1+1]$

EmpID	Name	Salary	Dept
101	Anita	35000	Sales
102	Bikash	28000	IT
103	Chandra	42000	Finance

i. Write a SQL code to create the above table (make EmpID the Primary key).

ii. Write a SQL code to display records of employees whose Salary is greater than 30000.

iii. Insert a new record (104, Deepa, 25000, HR).

12. Answer the following questions.

a. Write a Python program to input the radius of a circle and calculate its area and circumference using a function. $[1+1+2]$

b. Explain the concept of client-server architecture along with its major advantages. $[4]$

c. Define exception in Python. Explain the try, except, else and finally blocks with one suitable example. $[1+3]$

— **Best of Luck** —

Solutions — Practice Set 2

Group 'A' — Answer Key

1. What do you call the physical or logical arrangement of computers in a network? → **(b) Topology**
2. What is a foreign key used for in a database? → **(b) To establish and enforce a link between two tables**
3. Which Python function converts a string value into an integer? → **(a) int()**
4. What does an IP address represent? → **(b) A device's unique network identifier**
5. What type of argument has a pre-defined value that is used when no value is provided during a function call? → **(c) Default argument**
6. Which pandas function is used to read data from an Excel file? → **(b) read_excel()**
7. Which keyword is used to import an entire module in Python? → **(c) import**
8. Why is optical fiber preferred for long-distance, high-speed data transmission? → **(c) It offers high bandwidth and low signal loss**
9. Which Plotly Express function is used to create a pie chart? → **(b) px.pie()**
10. Which SQL command is used to retrieve records from a table? → **(b) SELECT**

Group 'B' — Answers

(a) Network topology is the arrangement or layout pattern in which computers and devices are interconnected in a network. Two types are Ring topology and Mesh topology.

(b) Write a Python turtle syntax that can draw a rectangle.

```
import turtle
t = turtle.Turtle()
for i in range(2):
    t.forward(100)
    t.right(90)
    t.forward(50)
    t.right(90)
```

(c) Data transmission speed is the rate at which data is transferred from one point to another over a network. It is measured in bits per second (bps), commonly expressed in Kbps, Mbps, or Gbps.

(d) A foreign key is a field in one table that refers to the primary key of another table. It is used to create and enforce a relationship (link) between two tables, maintaining referential integrity.

(e) DDL (Data Definition Language) consists of SQL commands used to define and modify the structure of database objects. Two commands: CREATE and ALTER.

(f) Guided media (wired media) is transmission media in which signals travel through a physical, solid medium. Examples: Twisted pair cable and Coaxial cable.

(g) 1) Wired media transmits signals through physical cables, while wireless media transmits signals through air using electromagnetic waves. 2) Wired media is generally more secure and less prone to interference, while wireless media offers greater mobility and easier installation.

(h) A signal is an electrical, electromagnetic, or optical representation of data that varies with time and is used to carry information from a sender to a receiver over a communication channel.

(i) Data cleaning is the process of detecting and correcting (or removing) inaccurate, incomplete, duplicate, or irrelevant data from a dataset to improve its quality before analysis.

(j) The global keyword is used inside a function to indicate that a variable refers to the one defined in the global scope, allowing the function to modify a global variable rather than creating a new local one.

Group 'C' — Answers

11. (a)

A database is an organized collection of related data stored in a structured way for easy access and management. A DBMS is software used to create, access, manage, and secure the data stored in a database (e.g., MySQL, PostgreSQL).

A primary key is a field (or set of fields) in a table that uniquely identifies each record and cannot contain NULL or duplicate values.

Example: in an Employee table, "EmpID" is set as the primary key because each employee has a distinct ID.

11. (b) Table: Employee

```
CREATE TABLE Employee (  
    EmpID INT PRIMARY KEY,  
    Name VARCHAR(30),  
    Salary INT,  
    Dept VARCHAR(20)  
);  
  
SELECT * FROM Employee WHERE Salary > 30000;  
  
INSERT INTO Employee VALUES (104, 'Deepa', 25000, 'HR');
```

12. (a)

```
import math  
  
def circle_calc(r):  
    area = math.pi * r * r  
    circumference = 2 * math.pi * r  
    print("Area =", round(area, 2))  
    print("Circumference =", round(circumference, 2))  
  
radius = float(input("Enter radius: "))  
circle_calc(radius)
```

12. (b)

Client-server architecture is a distributed computing model where client machines send requests over a network to a central server, which processes those requests and returns the required data or service. Web browsing, email, and online banking are common examples.

Major advantages: 1) Centralized control makes data management, security, and backup simpler. 2) Multiple clients can share the same resources (files, databases, printers) hosted on the server. 3) The system is scalable — servers can be upgraded or clients added independently. 4) Maintenance and software updates can be applied centrally on the server rather than on every machine.

12. (c)

An exception is an error that occurs during the execution of a program (at runtime) and disrupts the normal flow of the program's instructions, e.g., dividing by zero or accessing an invalid list index.

The try block holds the code that might raise an exception. The except block catches the exception and defines how to handle it. The else block executes only if the try block completes without raising an exception. The finally block always executes at the end, regardless of whether an exception occurred, and is normally used to release resources.

```
try:  
    marks = [70, 80, 90]  
    index = int(input("Enter index: "))  
    print("Marks:", marks[index])  
except IndexError:  
    print("Index out of range!")  
except ValueError:
```

```
        print("Please enter a valid integer!")
else:
    print("Value fetched successfully.")
finally:
    print("Program ended.")
```

Symbol No. _____

Sub code '3031'

Practice Question — Set 3

First Terminal Examination - 2083

Subject: Optional Computer Science

Class: X Group 'A'

Full Marks: 50 Time: 2 hrs

Group 'A'

Rewrite the correct option in your answer sheet. [10 × 1 = 10]

- 1. What is the term for the physical or logical arrangement of nodes in a network called?**
(a) Bandwidth (b) Protocol (c) Topology (d) Node
- 2. Which SQL key ensures that a column cannot have duplicate or NULL values and uniquely identifies each row?**
(a) Foreign key (b) Primary key (c) Candidate key (d) Composite key
- 3. Which Python statement is used to display multiple values separated by a comma?**
(a) print() (b) input() (c) format() (d) show()
- 4. What is the main function of a router in a network?**
(a) It stores files permanently (b) It forwards data packets between different networks (c) It converts AC power to DC (d) It scans documents
- 5. Which of the following is an example of a keyword argument in a Python function call?**
(a) func(5, 10) (b) func(a=5, b=10) (c) func(5) (d) def func():
- 6. Which parameter of to_csv() is used to specify the column separator in the output file?**
(a) sep (b) header (c) index (d) mode
- 7. Which statement is used to import a module with an alias name in Python?**
(a) import pandas (b) import pandas as pd (c) from pandas import * (d) include pandas
- 8. Why is Bluetooth preferred over LAN cables for connecting a wireless mouse to a laptop?**
(a) It uses fiber optic cables (b) It supports only wired devices (c) It enables short-range wireless connection between nearby devices (d) It requires an internet connection
- 9. Which Plotly Express function is used to create a scatter plot?**
(a) px.line() (b) px.bar() (c) px.scatter() (d) px.pie()
- 10. Which SQL command is used to add a new record into a table?**
(a) UPDATE (b) INSERT (c) DELETE (d) SELECT

Group 'B'

Write short answers to the following questions. [10 × 2 = 20]

- (a) What is meant by network topology? List its types.**
- (b) Write a Python turtle syntax that can draw a triangle.**
- (c) What is meant by protocol in networking? Give one example.**
- (d) What is a candidate key? How is it different from a primary key?**
- (e) Define DCL with any two commands.**
- (f) What is unguided (wireless) media? Give two examples.**
- (g) Differentiate between synchronous and asynchronous transmission with two points.**
- (h) What is attenuation in telecommunication?**

(i) What is the importance of data visualization?

(j) Define parameter and argument in a Python function with an example.

Group 'C'

Write long answers to the following questions. $[4 \times 5 = 20]$

11. Answer the following questions.

a. Define DBMS. Explain the difference between field and record with a suitable example. $[2+1+1]$

b. Consider the following table "Book": $[2+1+1]$

BookID	Title	Price	Genre
1	Python Basics	550	Programming
2	Nepali Sanskriti	400	Culture
3	World History	650	History

i. Write a SQL code to create the above table (make BookID the Primary key).

ii. Write a SQL code to display records of books whose Price is greater than 500.

iii. Insert a new record (4, 'Data Science 101', 720, 'Technology').

12. Answer the following questions.

a. Write a Python program to input the marks of three subjects and calculate the total and percentage using a function. $[1+1+2]$

b. Explain client-server architecture along with its major advantages. $[4]$

c. Define runtime error. Explain the try, except, else and finally blocks with one suitable example. $[1+3]$

— Best of Luck —

Solutions — Practice Set 3

Group 'A' — Answer Key

1. What is the term for the physical or logical arrangement of nodes in a network called? → **(c) Topology**
2. Which SQL key ensures that a column cannot have duplicate or NULL values and uniquely identifies each row? → **(b) Primary key**
3. Which Python statement is used to display multiple values separated by a comma? → **(a) print()**
4. What is the main function of a router in a network? → **(b) It forwards data packets between different networks**
5. Which of the following is an example of a keyword argument in a Python function call? → **(b) func(a=5, b=10)**
6. Which parameter of `to_csv()` is used to specify the column separator in the output file? → **(a) sep**
7. Which statement is used to import a module with an alias name in Python? → **(b) import pandas as pd**
8. Why is Bluetooth preferred over LAN cables for connecting a wireless mouse to a laptop? → **(c) It enables short-range wireless connection between nearby devices**
9. Which Plotly Express function is used to create a scatter plot? → **(c) px.scatter()**
10. Which SQL command is used to add a new record into a table? → **(b) INSERT**

Group 'B' — Answers

(a) Network topology refers to the pattern in which nodes (computers and devices) of a network are connected to each other, either physically or logically. Common types include Bus, Star, Ring, Mesh, and Tree topology.

(b) Write a Python turtle syntax that can draw a triangle.

```
import turtle
t = turtle.Turtle()
for i in range(3):
    t.forward(100)
    t.left(120)
```

(c) A protocol is a standardized set of rules that determines how data is formatted, transmitted, and received between devices in a network so that they can communicate correctly. Example: FTP (File Transfer Protocol).

(d) A candidate key is any column or set of columns that can qualify to uniquely identify records in a table. A table may have several candidate keys, but only one of them is chosen as the primary key, while the rest remain as alternate keys.

(e) DCL (Data Control Language) consists of SQL commands used to control access and permissions to data in a database. Two commands: GRANT and REVOKE.

(f) Unguided media is transmission media that carries signals through open space without a physical conductor, using electromagnetic waves. Examples: Radio waves and Microwaves.

(g) 1) In synchronous transmission, data is sent in a continuous stream synchronized by a shared clock signal, whereas in asynchronous transmission, data is sent one character at a time with start and stop bits. 2) Synchronous transmission is faster and more efficient for large data, while asynchronous transmission is simpler and cheaper but slower.

(h) Attenuation is the gradual loss or weakening of signal strength as it travels through a transmission medium over distance.

(i) Data visualization presents data in graphical or pictorial form, making patterns, trends, and outliers easier to understand quickly, which supports faster and more informed decision-making.

(j) A parameter is a variable listed in a function's definition (e.g., 'name' in `def greet(name):`), while an argument is the actual value passed to the function when it is called (e.g., `greet("Ram")` — here "Ram" is the argument).

Group 'C' — Answers

11. (a)

A DBMS (Database Management System) is software that enables users to define, create, maintain, and control access to a database, allowing efficient storage, retrieval, and manipulation of data (e.g., MySQL, SQLite).

A field is a single column in a table representing one attribute of the data, e.g., "Price" in a Book table.

A record is a complete row in a table containing all field values for one entity, e.g., (1, 'Python Basics', 550, 'Programming') is one record.

11. (b) Table: Book

```
CREATE TABLE Book (  
    BookID INT PRIMARY KEY,  
    Title VARCHAR(50),  
    Price INT,  
    Genre VARCHAR(30)  
);  
  
SELECT * FROM Book WHERE Price > 500;  
  
INSERT INTO Book VALUES (4, 'Data Science 101', 720, 'Technology');
```

12. (a)

```
def calculate_result(m1, m2, m3):  
    total = m1 + m2 + m3  
    percentage = total / 3  
    print("Total =", total)  
    print("Percentage =", round(percentage, 2))  
  
sub1 = float(input("Enter marks of subject 1: "))  
sub2 = float(input("Enter marks of subject 2: "))  
sub3 = float(input("Enter marks of subject 3: "))  
calculate_result(sub1, sub2, sub3)
```

12. (b)

Client-server architecture is a network model in which tasks are divided between service providers (servers) and service requesters (clients). The client sends a request over the network, and the server processes it and returns the result — for example, a web browser (client) requesting a webpage from a web server.

Major advantages: 1) Centralized storage and management make it easier to secure and back up data. 2) Shared access to resources such as files, databases, and printers reduces duplication. 3) The architecture is scalable, allowing new clients to be added with minimal disruption. 4) Centralized administration simplifies software updates and troubleshooting.

12. (c)

A runtime error is an error that occurs while a program is being executed (even though its syntax is correct), causing it to behave unexpectedly or stop abruptly, e.g., dividing a number by zero.

The try block contains the code that might cause an error. The except block catches and handles that error so the program does not crash. The else block runs only when the try block executes successfully without any error. The finally block always runs at the end, whether or not an error occurred, and is generally used for clean-up tasks such as closing a file.

```
try:  
    a = int(input("Enter dividend: "))  
    b = int(input("Enter divisor: "))
```

```
    result = a / b
except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")
except ValueError:
    print("Error: Please enter valid integers.")
else:
    print("Result =", result)
finally:
    print("Calculation attempt finished.")
```